STA2002: Probability and Statistics II
Maximum Likelihood Estimation II

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Introduction

- In this lecture we will continue our discussion on the maximum likelihood estimation method. We will also introduce the notion of unbiasedness, and introduce another method of parameter estimation called the method of moments (mom).
- **Suggested reading**: Chapter 6.4 in the book. (MLE & mom)

Quality of statistics. e.g. unbiasness ; consistency

$\bar{X} \xrightarrow{P} \mu$

$S^2 \xrightarrow{P} \sigma^2$

$\bar{X}$ a statistic $T(X_1, \ldots X_n)$ is random variable.

Example: $N(\theta_1, \theta_2)$

Example 1

This is Example 6.4-3 in the book. Suppose that $X_1, \ldots X_n \sim N(\theta_1, \theta_2)$. The unknown parameter is two-dimensional $\theta = (\theta_1, \theta_2)$, and the pdf is given by

$$f(x;\theta) = \frac{1}{\sqrt{2\pi\theta_2}} \exp\left[-\frac{(x-\theta_1)^2}{2\theta_2}\right], \quad -\infty < x < \infty.$$

$$\theta \in \Omega = \{(\theta_1, \theta_2) : -\infty < \theta_1 < \infty, 0 < \theta_2 < \infty\}.$$

The likelihood function is

$$L(\theta_1, \theta_2) = \prod_{i=1}^{n} f(x_i;\theta) = \left(\frac{1}{\sqrt{2\pi\theta_2}}\right)^n \exp\left[\frac{-\sum_{i=1}^{n}(x_i-\theta_1)^2}{2\theta_2}\right].$$

The log-likelihood function is

$$l(\theta_1, \theta_2) = -\frac{n}{2}\ln(2\pi\theta_2) - \frac{\sum_{i=1}^{n}(x_i-\theta_1)^2}{2\theta_2}.$$

Example: $N(\theta_1, \theta_2)$ (Cont'd)

- Taking the partial derivatives of $l(\theta_1, \theta_2)$ with respect to θ_1 and θ_2 and set them to zero give

$$\frac{\partial}{\partial \theta_1} l(\theta_1, \theta_2) = \frac{1}{\theta_2} \sum_{i=1}^{n} (x_i - \theta_1) = 0$$

$$\frac{\partial}{\partial \theta_2} l(\theta_1, \theta_2) = \frac{-n}{2\theta_2} + \frac{1}{2\theta_2^2} \sum_{i=1}^{n} (x_i - \theta_1)^2 = 0$$

- The solutions are

$$\theta_1 = \frac{\sum_{i=1}^{n} x_i}{n} = \bar{x},$$

$$\theta_2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2 .$$

Example: $N(\theta_1, \theta_2)$ (Cont'd)

- We check that this solution is indeed the global maximum by the second derivative test.
- As a result, the mle are

$$\hat{\theta}_1 = \frac{\sum_{i=1}^n X_i}{n} = \overline{X},$$
$$\hat{\theta}_2 = \frac{1}{n}\sum_{i=1}^n \left(X_i - \overline{X}\right)^2.$$

- $\overline{X}$ is the sample mean while $\frac{1}{n}\sum_{i=1}^n \left(X_i - \overline{X}\right)^2$ is called the variance of the empirical distribution.

Unbiased Estimator

Definition 2

An estimator $u(X_1, X_2, \ldots, X_n)$ of θ is an unbiased estimator of θ if

$$\mathbb{E}(u(X_1, X_2, \ldots, X_n)) = \theta.$$

Otherwise, $u(X_1, X_2, \ldots, X_n)$ is called a biased estimator.

Example 3 (Example: $N(\theta_1, \theta_2)$ continued)

This is Example 6.4-5 in the book. Suppose that $X_1, \ldots X_n \sim N(\theta_1, \theta_2)$. The unknown parameter is two-dimensional $\theta = (\theta_1, \theta_2)$.

Task: To check that $\hat{\theta}_1$ is an unbiased estimator of θ_1, and $\hat{\theta}_2$ is a biased estimator of θ_2.

Solution

- Recall that the mle are,

$$\hat{\theta}_1 = \frac{\sum_{i=1}^{n} X_i}{n} = \overline{X},$$
$$\hat{\theta}_2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2 = \frac{n-1}{n}\left(\frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2\right) = \frac{n-1}{n}S^2.$$

- Recall also that,

$$\overline{X} \sim N(\theta_1, \theta_2/n),$$
$$\frac{(n-1)S^2}{\theta_2} \sim \chi^2(n-1).$$

Solution (Cont'd)

- As a result,

$$\mathbb{E}(\hat{\theta}_1) = \mathbb{E}(\overline{X}) = \theta_1,$$

$$\mathbb{E}(S^2) = \frac{\theta_2}{n-1}\mathbb{E}\left(\frac{(n-1)S^2}{\theta_2}\right) = \theta_2.$$

$\frac{(n-1)S^2}{\theta_2}$ 卡方(n-1)

- That is, $\hat{\theta}_1$ is an unbiased estimator of θ_1, and S^2 is an unbiased estimator of θ_2.
- For $\hat{\theta}_2$, however, since

$$\mathbb{E}(\hat{\theta}_2) = \frac{n-1}{n}\mathbb{E}(S^2) = \frac{n-1}{n}\theta_2,$$

$\hat{\theta}_2$ is a biased estimator of θ_2. □

Method of moments (mom)

- Using the maximum likelihood estimation method requires maximizing the likelihood function $L(\theta)$.
- At times it can be hard to give an explicit formula for the maximum value, and so numerical optimization methods are required.
- Another method to derive a point estimator is called the method of moments (mom).

▷ What's the problem that we want to solve?

▷ The key idea behind the technique (mom)

▷ The calculation procedure.

▷ The properties of Mom

▷ Comparison with MLE

Method of moments (mom)

- Suppose that the unknown parameter $\theta = (\theta_1, \ldots, \theta_k)$ is k-dimensional. For $1 \leq j \leq k$, denote the jth moment to be

$$\alpha_j(\theta) = \mathbb{E}_\theta(X^j).$$

- Denote the jth sample moment to be

$$\hat{\alpha}_j = \frac{1}{n}\sum_{i=1}^{n} X_i^j.$$

theoretical moment
population ~

sample ~
emprical moment

sample moment is sample mean.

theoratical moment is expectation of X^j.

Method of moments (mom)

Definition 4 (method of moments estimator)

The method of moments estimator $\widetilde{\theta}$ is defined to be the value of θ such that

[theoratical m~ = sample moment] key idea

$$\alpha_1(\theta) = \hat{\alpha}_1$$
$$\alpha_2(\theta) = \hat{\alpha}_2$$
$$\vdots \quad \vdots \quad \vdots$$
$$\alpha_k(\theta) = \hat{\alpha}_k.$$

It defines a system of k equations with k unknowns.

Example: Gamma(θ_1, θ_2)

$\mu = \theta_1 \theta_2$

$\sigma^2 = \theta_1 \theta_2^2$

Example 5

Suppose that $X_1, \ldots X_n \sim \text{Gamma}(\theta_1, \theta_2)$. *The unknown parameter is two-dimensional* $\theta = (\theta_1, \theta_2)$, *and its first two moments are*

$$\alpha_1(\theta_1, \theta_2) = \mathbb{E}_\theta(X_1) = \theta_1 \theta_2,$$
$$\alpha_2(\theta_1, \theta_2) = \mathbb{E}_\theta(X_1^2) = \theta_1 \theta_2^2 + \theta_1^2 \theta_2^2.$$

The first two sample moments are

$$\hat{\alpha}_1 = \frac{1}{n} \sum_{i=1}^{n} X_i,$$
$$\hat{\alpha}_2 = \frac{1}{n} \sum_{i=1}^{n} X_i^2.$$

Example: Gamma(θ_1, θ_2) (Cont'd)

- To carry out the method of moments, we set the first moment to equal to the first sample moment, and the second moment to equal to the second sample moment, that is,

$$\theta_1\theta_2 = \frac{1}{n}\sum_{i=1}^{n} X_i = \overline{X}, \tag{1}$$

$$\theta_1\theta_2^2 + \theta_1^2\theta_2^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2. \tag{2}$$

- Let

$$V := \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i^2 - 2\overline{X}X_i + \overline{X}^2\right) = \frac{1}{n}\left(\sum_{i=1}^{n} X_i^2\right) - \overline{X}^2.$$

Example: Gamma(θ_1, θ_2) (Cont'd)

- Plugging (1) into (2) yields

$$\overline{X}\theta_2 + \overline{X}^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2.$$

- Rearranging gives

$$\theta_2 = \frac{V}{\overline{X}}.$$

  Substituting this θ_2 back into (1) gives

$$\theta_1 = \frac{\overline{X}^2}{V}.$$

- The method of moments estimators are therefore

$$\widetilde{\theta}_1 = \frac{\overline{X}^2}{V},$$
$$\widetilde{\theta}_2 = \frac{V}{\overline{X}}.$$

Example: Uniform$(0, \theta)$

Example 6

Suppose that $X_1, \ldots X_n$ are uniformly distributed on the interval $[0, \theta]$.
Recall that the maximum likelihood estimator $\hat{\theta}$ is

$$\hat{\theta} = \max\{X_1, \ldots, X_n\}.$$

- Using the method of moments, we set the first moment to equal to the first sample moment, that is,

$$\frac{\theta}{2} = \frac{1}{n}\sum_{i=1}^{n} X_i = \overline{X}.$$

Example: Uniform$(0, \theta)$

- Therefore, the mom estimator is

$$\widetilde{\theta} = 2\overline{X}.$$

MLE ↓ $\hat{\theta}$ mom ↓ $\widetilde{\theta}$

- Note that $\hat{\theta} \neq \widetilde{\theta}$. Also, $\widetilde{\theta}$ may not be a good estimator if $2\overline{X} < \max\{X_1, \ldots, X_n\}$. □